A SEPARATED MORPHISM THAT IS LOCALLY AN OPEN IMMERSION AND IS GENERICALLY INJECTIVE ON IRREDUCIBLE COMPONENTS IS AN OPEN IMMERSION

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof cites no external \ref inside the counted span, so the assembly needs no external-reference stubs.

Allowed standard prerequisites (editorial).

- Separatedness and uniqueness of limits / specialization lifting. Stacks Project, Schemes (separated morphisms and specialization to a point). Used to conclude x = x' once their generic specializations agree over the same point of Y.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let X^0 denote the set of generic points of irreducible components of X. If*

(1) *f is separated,*
(2) *there is an open covering $X = \bigcup U_i$ such that $f|_{U_i} : U_i \to Y$ is an open immersion, and*
(3) *if $\xi, \xi' \in X^0$, $\xi \neq \xi'$, then $f(\xi) \neq f(\xi')$,*

then f is an open immersion.

Proof. Suppose that $y = f(x) = f(x')$. Pick a specialization $y_0 \rightsquigarrow y$ where y_0 is a generic point of an irreducible component of Y. Since f is locally on the source an isomorphism we can pick specializations $x_0 \rightsquigarrow x$ and $x_0' \rightsquigarrow x'$ mapping to $y_0 \rightsquigarrow y$. Note that $x_0, x_0' \in X^0$. Hence $x_0 = x_0'$ by assumption (3). As f is separated we conclude that $x = x'$. Thus f is an open immersion. □

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